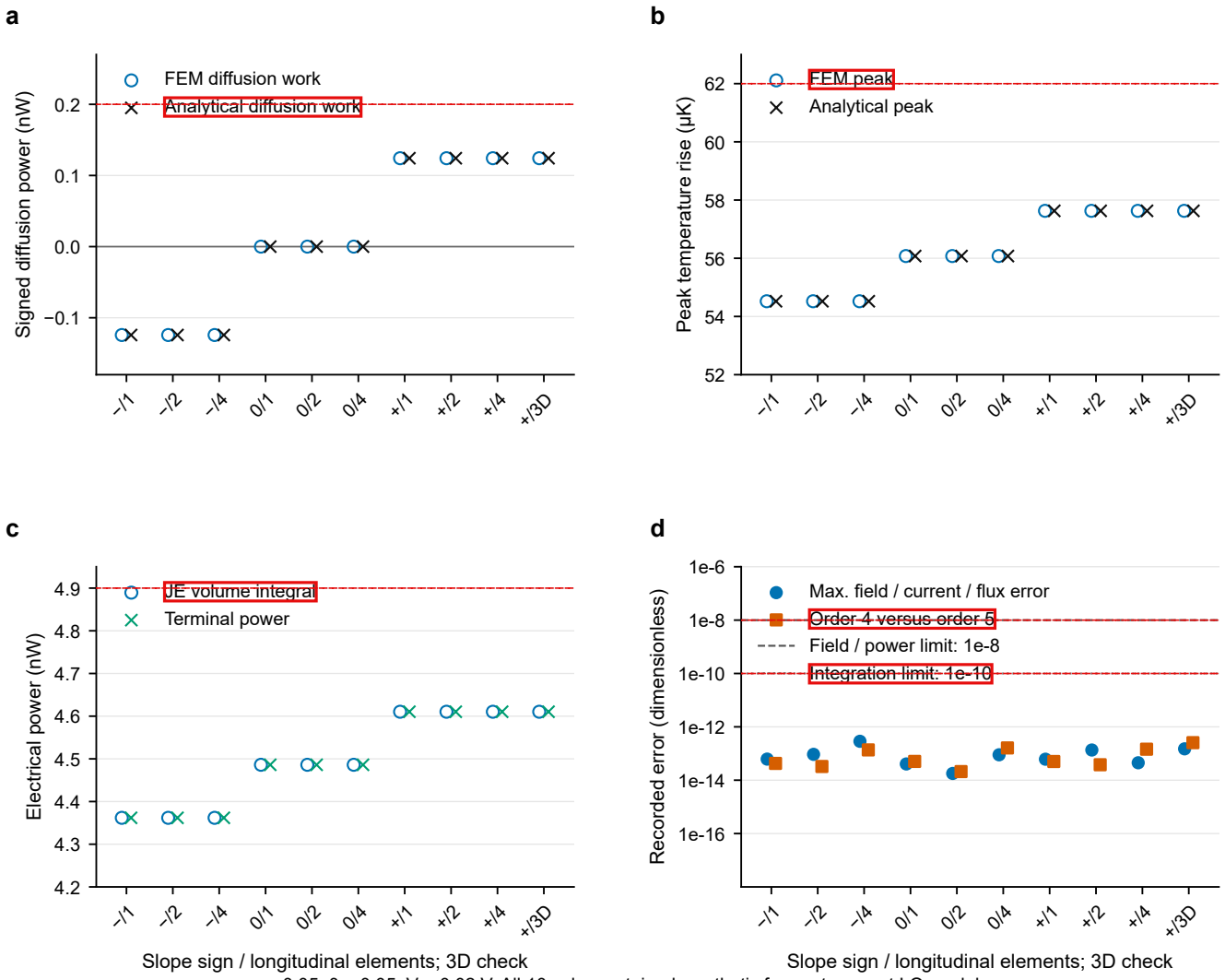

S2 | Signed diffusion-work benchmark (JEGRAD)
a
Signed diffusion power (nW)
FEM diffusion work
Analytical diffusion work
0.2
0.1
0.0
−0.1
b
Peak temperature rise (μK)
FEM peak
Analytical peak
62
60
58
56
54
52
c
Electrical power (nW)
JE volume integral
Terminal power
4.9
4.8
4.7
4.6
4.5
4.4
4.3
4.2
d
Recorded error (dimensionless)
Max. field / current / flux error
Order 4 versus order 5
Field / power limit: 1e-8
Integration limit: 1e-10
1e-6
1e-8
1e-10
1e-12
1e-14
1e-16
−/1
−/2
−/4
0/1
0/2
0/4
+/1
+/2
+/4
+/3D
Slope sign / longitudinal elements; 3D check
Slope sign / longitudinal elements; 3D check
s = −0.05, 0, +0.05; V = 0.02 V. All 10 solves retained; synthetic frozen-transport LQ model.